

Equilibrium Monte Carlo simulation of the 2D Ising model

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Molecular Modeling - Monte Carlo Methods

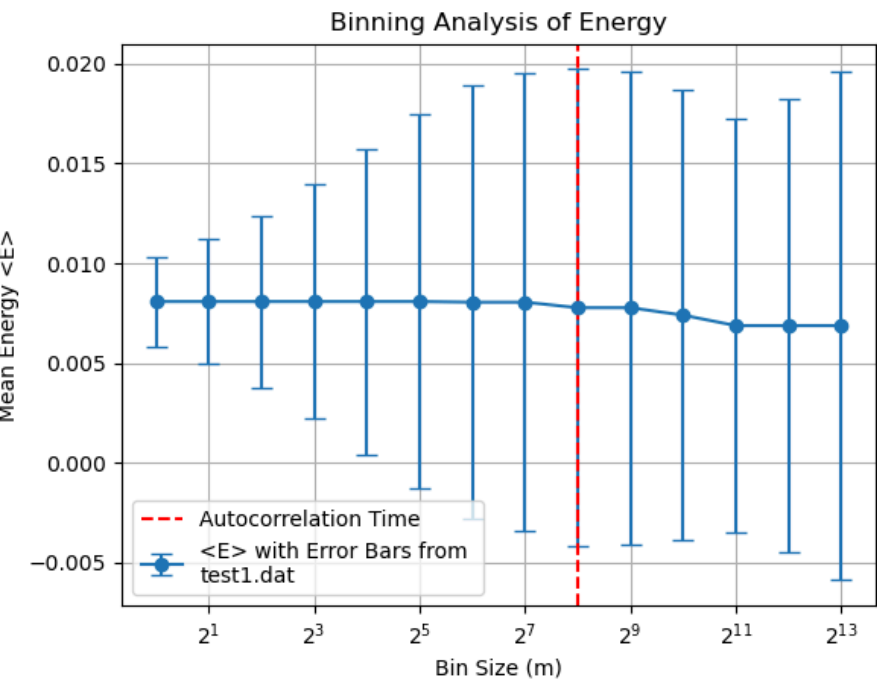
Required Task 2 - Monte Carlo Speed

Using numba just-in-time compilation, the code runs at the following rate:

```
raw
L=100, T=2.6, MCS=1000000
Time elapsed: 276.3268 s
Speed: 3.62e+07 updates/second
```

Required Task 3 - Binning & $\langle E \rangle / L^2$

test1.dat

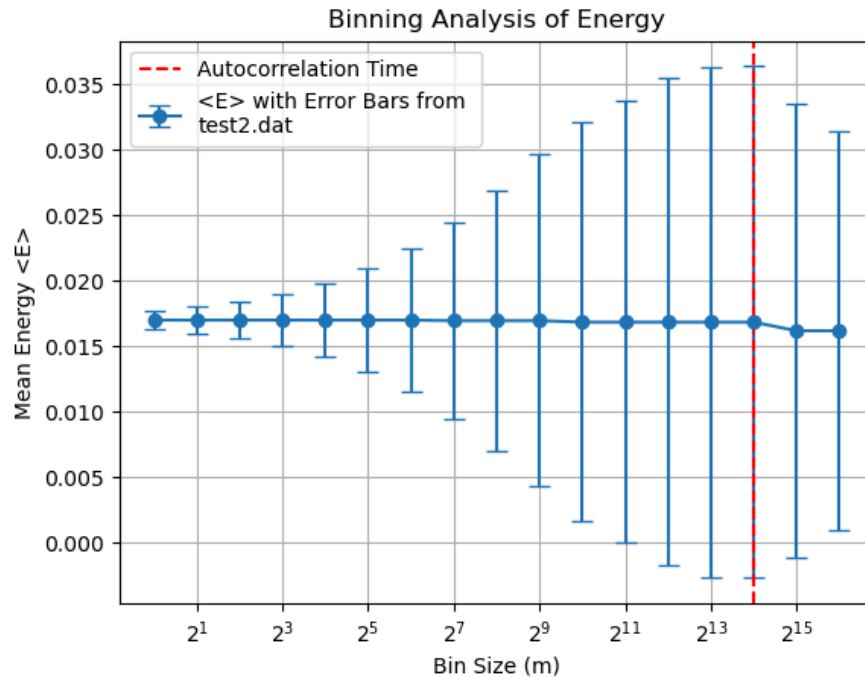


```
raw
Energy binning (E):
m      <E>      Error
1      0.008083  2.251117e-03
```

2	0.008083	3.130872e-03
4	0.008083	4.320143e-03
8	0.008083	5.854446e-03
16	0.008083	7.653067e-03
32	0.008083	9.382085e-03
64	0.008051	1.085487e-02
128	0.008051	1.146997e-02
256	0.007773	1.196137e-02
512	0.007773	1.185384e-02
1024	0.007399	1.126752e-02
2048	0.006875	1.038942e-02
4096	0.006875	1.135564e-02
8192	0.006875	1.276215e-02

Autocorrelation time for test1.dat: 256

test2.dat



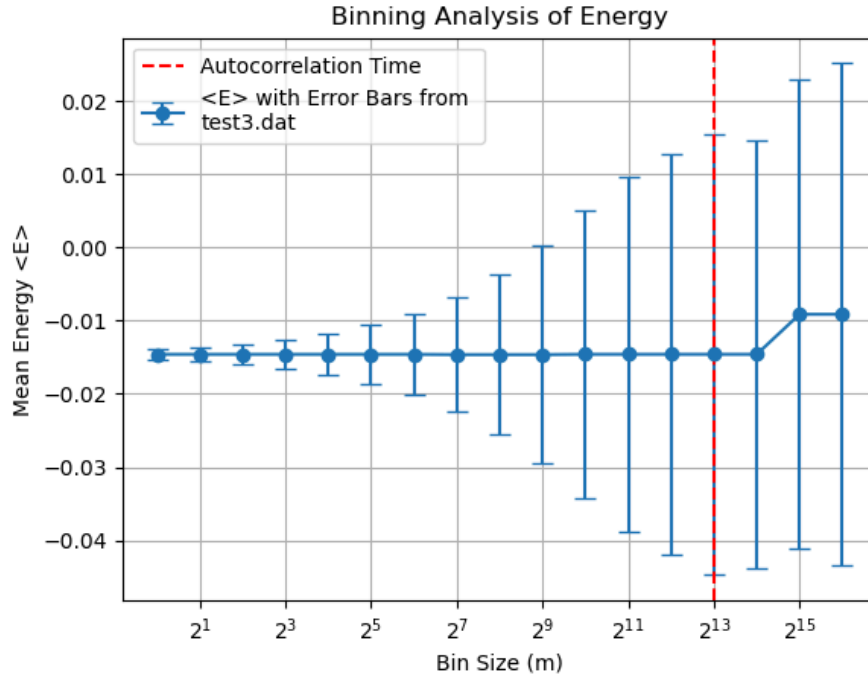
raw

Energy binning (E):

m	<E>	Error
1	0.017005	7.061275e-04
2	0.017005	9.978172e-04
4	0.017005	1.409438e-03
8	0.017005	1.988895e-03
16	0.017005	2.800694e-03
32	0.017005	3.927642e-03
64	0.017005	5.462279e-03
128	0.016964	7.479595e-03
256	0.016964	9.950295e-03
512	0.016964	1.268957e-02
1024	0.016849	1.523740e-02
2048	0.016849	1.687099e-02
4096	0.016849	1.860897e-02
8192	0.016849	1.944723e-02
16384	0.016849	1.954591e-02
32768	0.016193	1.737243e-02
65536	0.016193	1.525114e-02

Autocorrelation time for test2.dat: 16384

test3.dat



raw

Energy binning (E):

m	<E>	Error
1	-0.014588	7.061111e-04
2	-0.014588	9.983879e-04
4	-0.014588	1.411500e-03
8	-0.014588	1.995045e-03
16	-0.014588	2.818348e-03
32	-0.014588	3.977088e-03
64	-0.014588	5.599653e-03
128	-0.014629	7.851898e-03
256	-0.014629	1.091681e-02
512	-0.014629	1.489937e-02
1024	-0.014584	1.963385e-02
2048	-0.014584	2.426555e-02
4096	-0.014584	2.741836e-02
8192	-0.014584	3.017616e-02
16384	-0.014584	2.925151e-02
32768	-0.009114	3.210422e-02
65536	-0.009114	3.432141e-02

Autocorrelation time for test3.dat: 8192

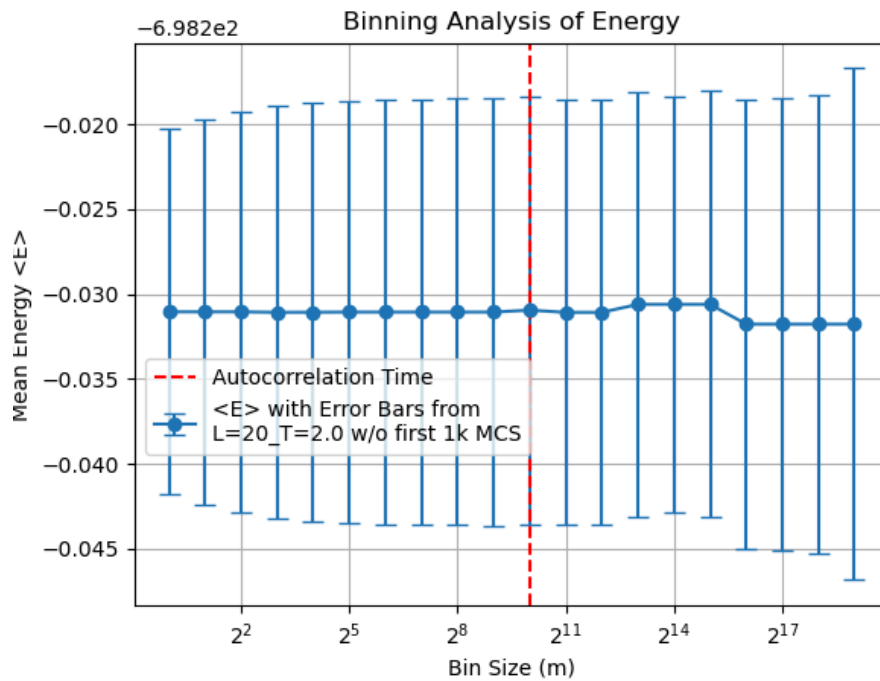
Required Task 4 - $\langle E \rangle / L^2$ with Statistical Error

Agreement

- When considering statistical error, results are within exact value given by Ferdinand and Fisher $\langle E \rangle / L^2 = -1.7455571250$

Parameters:

- $L = 20$
- $T = 2.0$
- $n_{meas} = 10$
- $n_{MCS} = 10^8$



raw

Energy binning (E):

m	$\langle E \rangle$	Error	$\langle E \rangle / L^2$	within exact range?
1	-698.231038	1.076861e-02	-1.745577595775958	yes
2	-698.231038	1.132774e-02	-1.745577595775958	yes
4	-698.231038	1.180775e-02	-1.745577595775958	yes
8	-698.231068	1.215100e-02	-1.745577669007758	yes
16	-698.231062	1.234180e-02	-1.745577654469730	yes
32	-698.231052	1.245284e-02	-1.745577630393669	yes
64	-698.231052	1.251481e-02	-1.745577630393669	yes
128	-698.231052	1.254391e-02	-1.745577630393669	yes
256	-698.231052	1.255955e-02	-1.745577630393669	yes
512	-698.231052	1.259953e-02	-1.745577630393669	yes
1024	-698.230940	1.260743e-02	-1.745577349950397	yes
2048	-698.231069	1.251124e-02	-1.745577673124808	yes
4096	-698.231069	1.249446e-02	-1.745577673124808	yes
8192	-698.230598	1.249915e-02	-1.745576494060579	yes
16384	-698.230598	1.223555e-02	-1.745576494060579	yes
32768	-698.230598	1.254956e-02	-1.745576494060579	yes
65536	-698.231764	1.320189e-02	-1.745579410352205	yes
131072	-698.231764	1.330383e-02	-1.745579410352205	yes
262144	-698.231764	1.352515e-02	-1.745579410352205	yes
524288	-698.231764	1.507062e-02	-1.745579410352205	yes

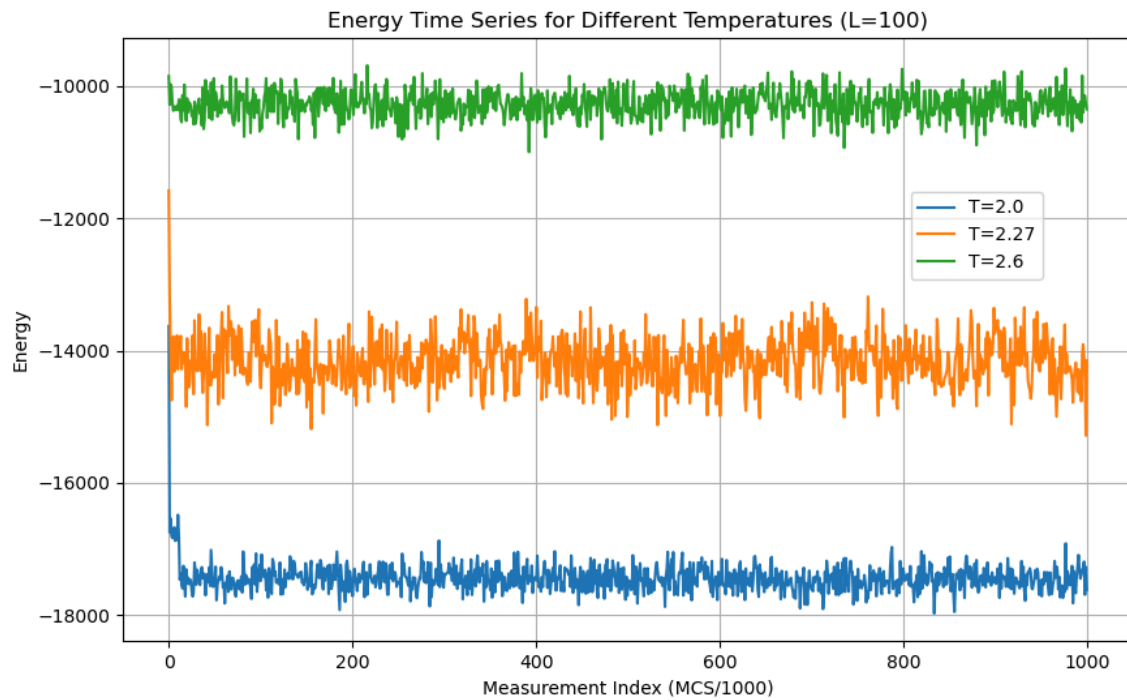
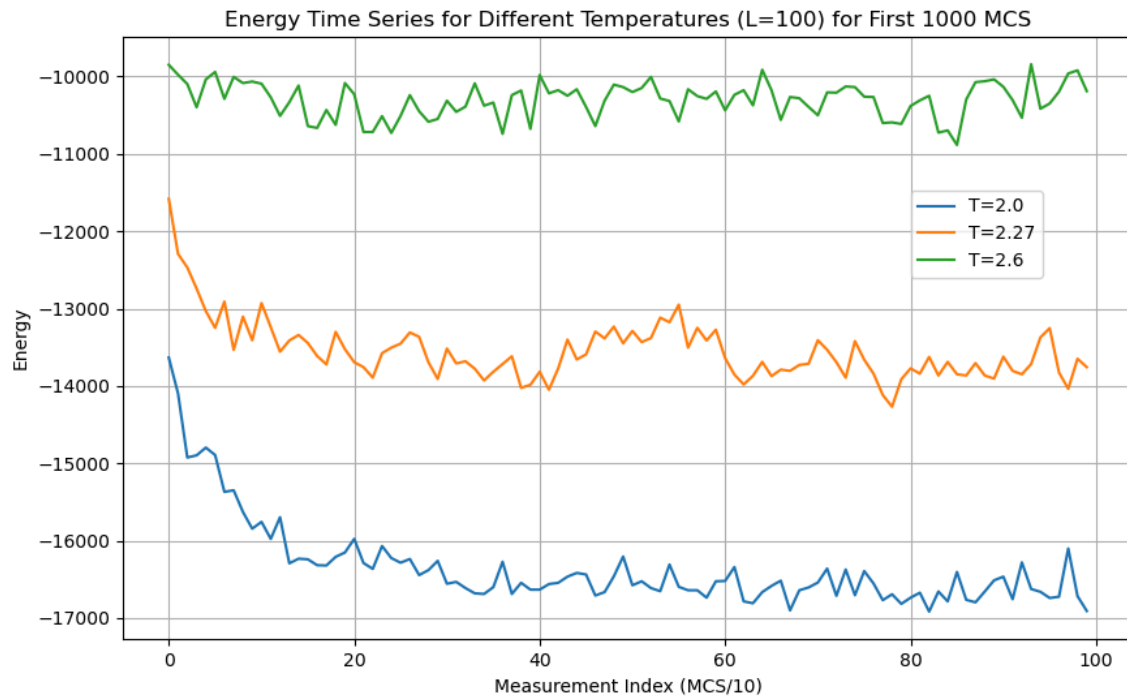
Required Task 6 - E & M Time Series at $T=2.0, 2.27, 2.6$

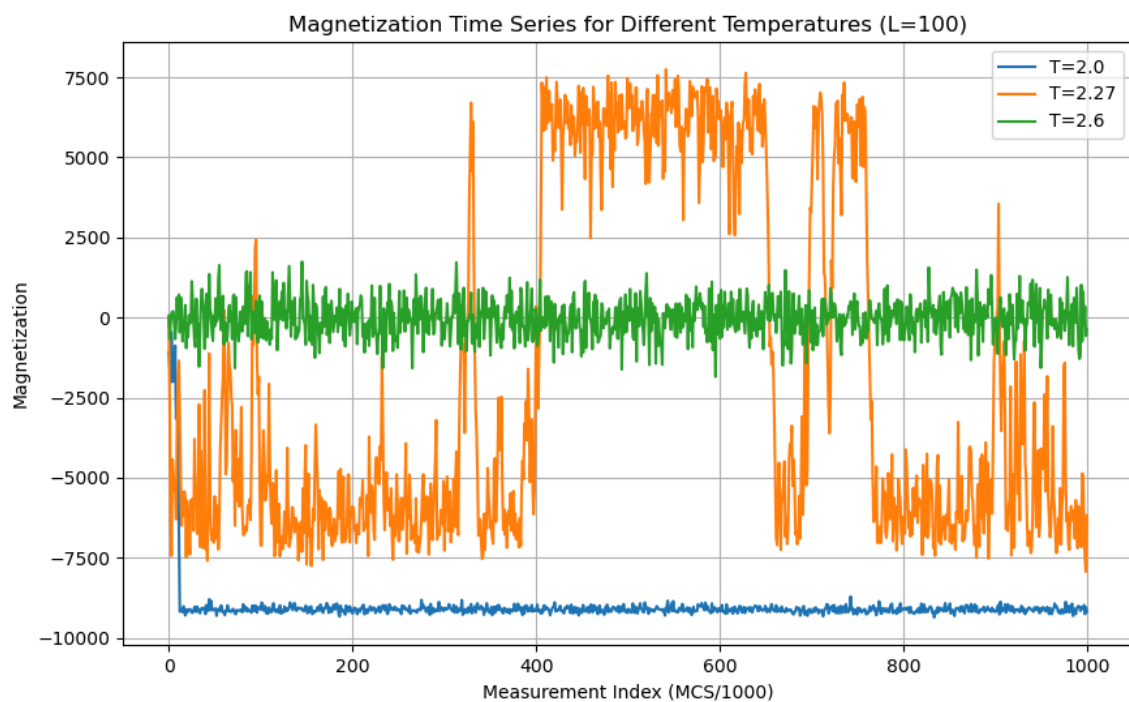
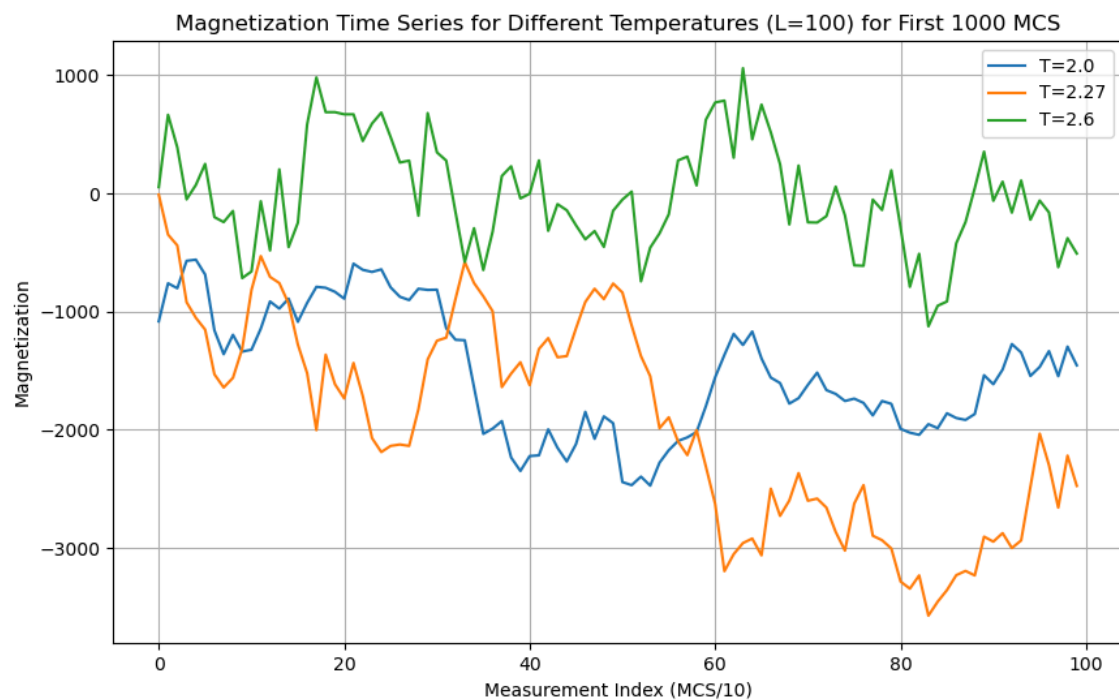
Interpreting $L=100$ Graphs

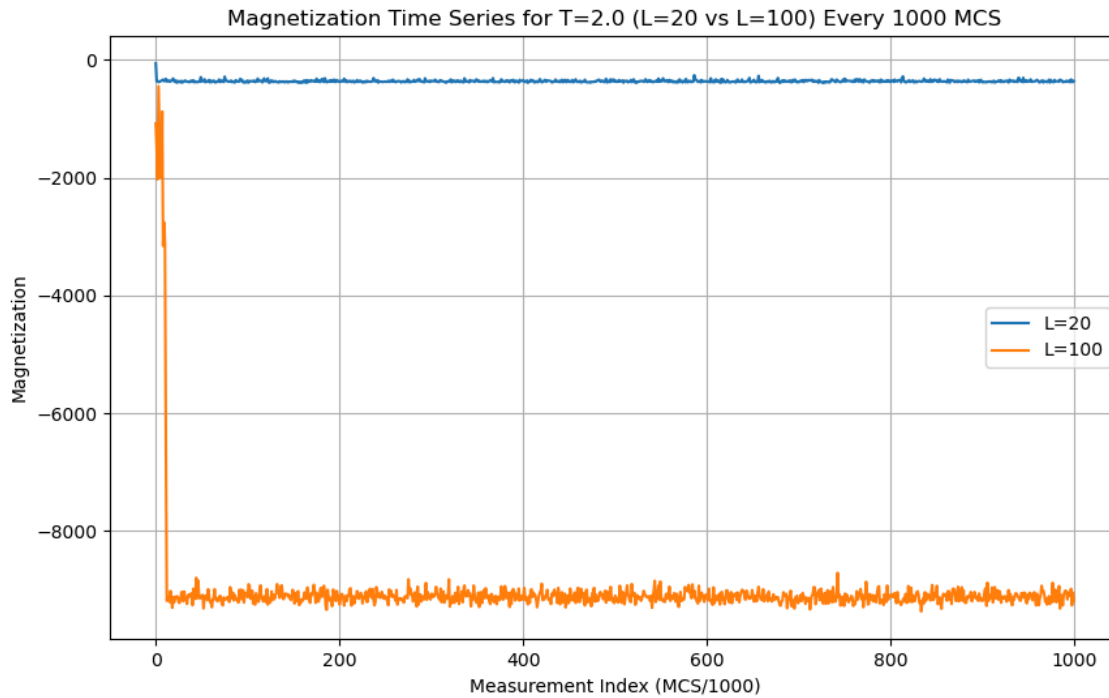
- Following an initial equilibration period, the energy E at each temperature is largely constant, though there are ~8-10% fluctuations. Higher fluctuations (about double) are seen at $T=2.27$, the critical temperature.
- Magnetization:
 - At $T=2.0$, the system is in the ferromagnetic phase and once equilibrated M stays constant around -9200 .
 - At $T=2.6$, it's in the paramagnetic phase. Here, M fluctuates around 0 and with higher variance than the ferromagnetic phase.
 - At $T=2.27$, the system is at the critical temperature where phase transition occurs. Here, M fluctuates with the highest variance (between ± 7500) and even changes sign (phase transition).

Comparing M for $L=20$ vs $L=100$ at $T=2.0$

- Both systems are in the ferromagnetic phase at $T=2.0$, but the $L=100$ system has a lower magnetization than the $L=20$ system. This is because the $L=100$ system is 25x larger (10000 vs 400), though each system shows the same *proportional* magnetization ($\sim 92\%$) to its respective system size.



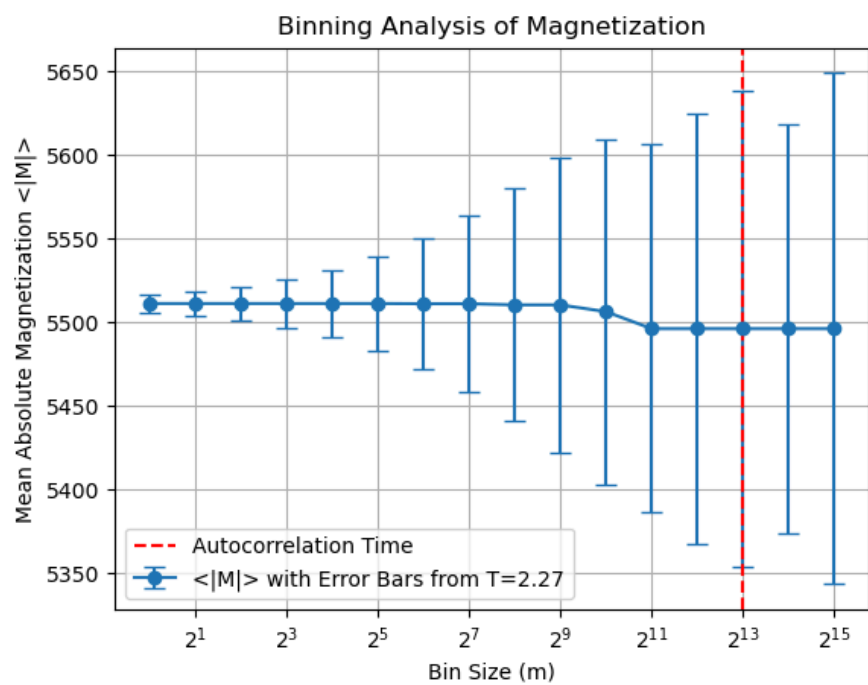
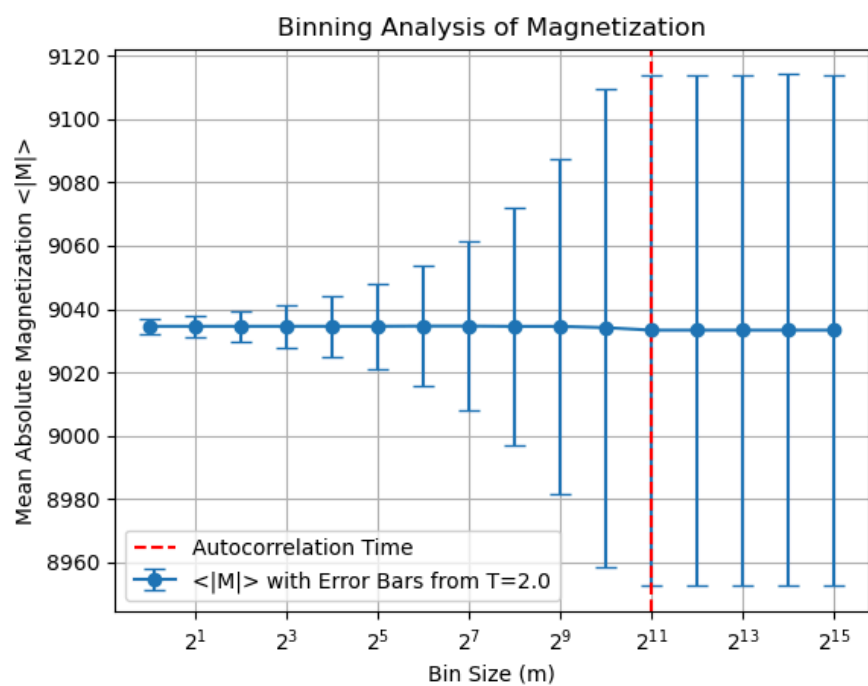


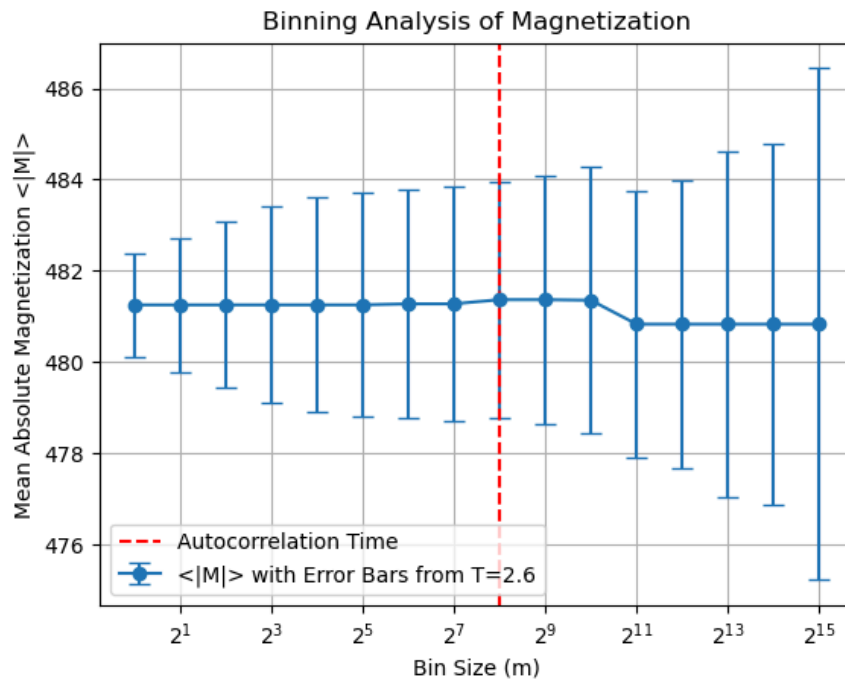


Required Task 7 - Plotting $\langle E \rangle$ & $\langle |M| \rangle$ with Statistical Error

Interpretation

- What we can observe in these three plots is how the estimated statistical error for energy and magnetization varies with the bin size used in the binning analysis. As we increase the bin size, the error tends to stabilize, indicating that we have reached a bin size large enough to capture the temporal correlations in the data.
- This is especially evident near the critical temperature ($T = 2.27$), where fluctuations are more pronounced and the autocorrelation time is longer. Below the critical temperature, the system exhibits more ordered behavior, which is reflected in smaller errors and a faster stabilization of the error with bin size. Above the critical temperature, the system is in a disordered phase, and due to the rapid decay of correlations, the error tends to stabilize even faster



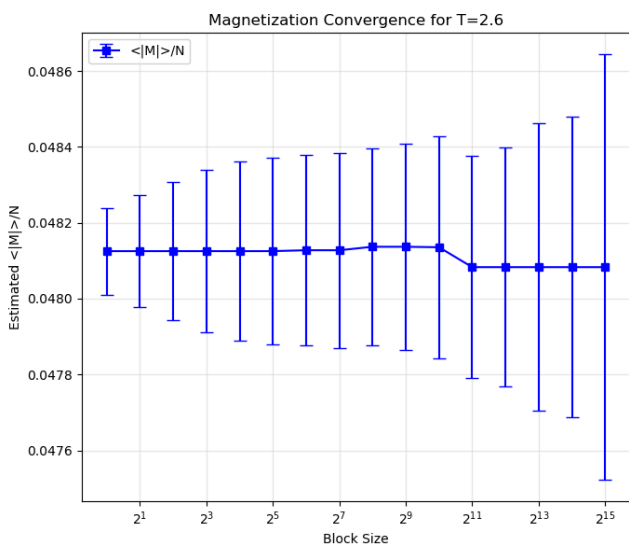
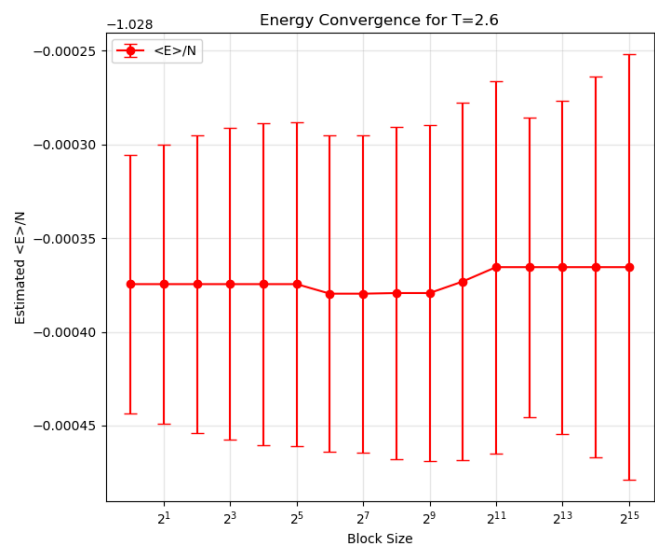
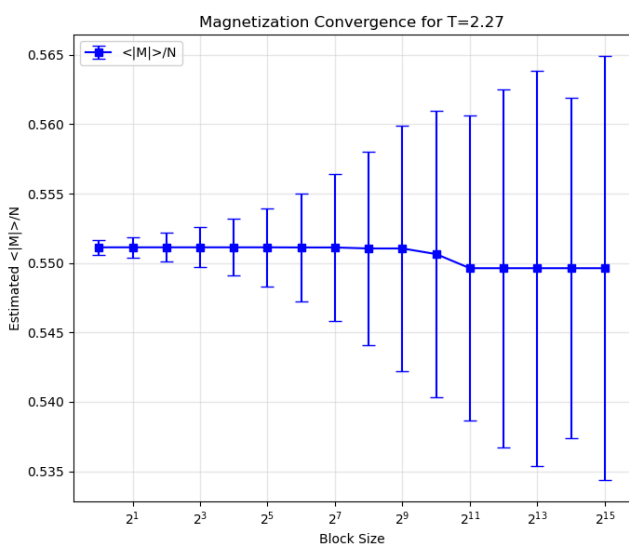
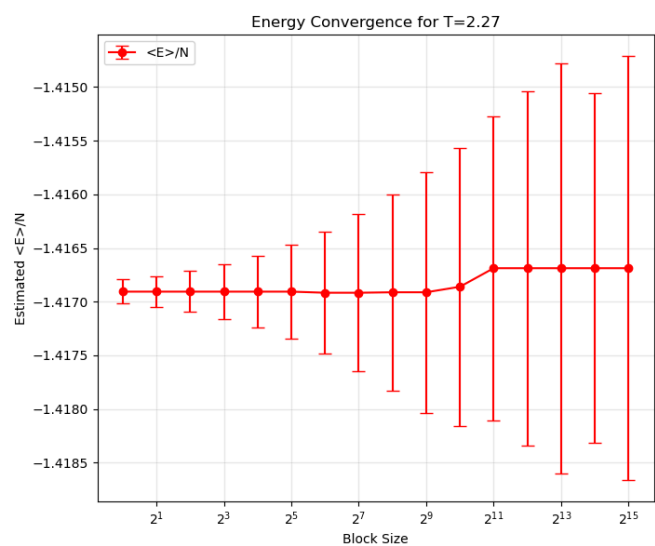
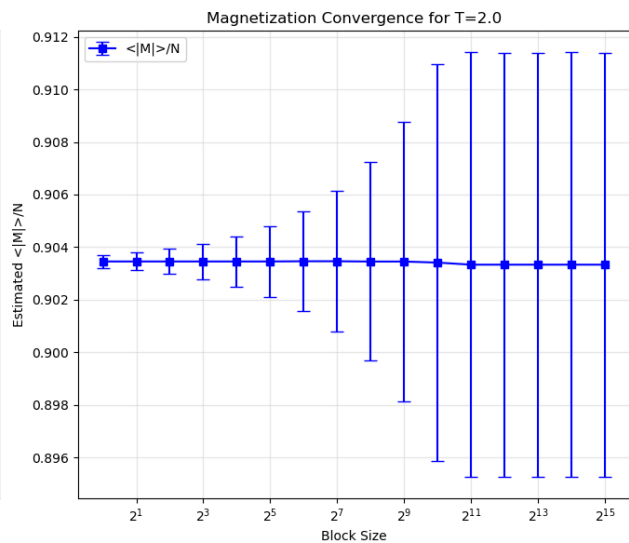
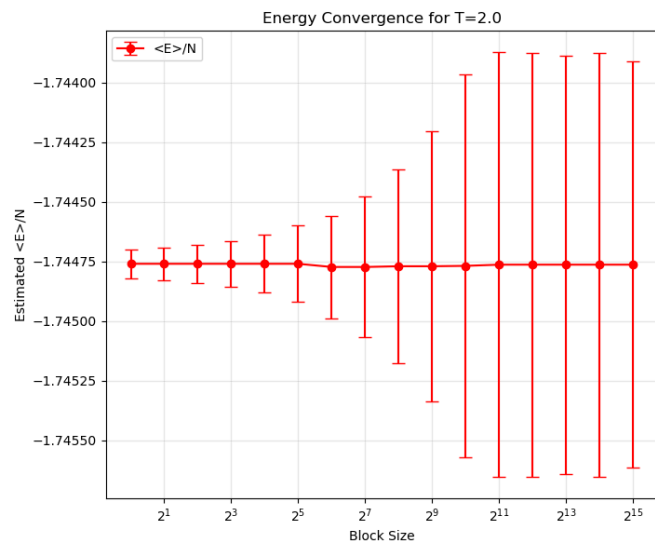


Required Task 8 - Plotting $\langle E \rangle / N$ and $\langle |M| \rangle / N$ with Statistical Error as a Function of T

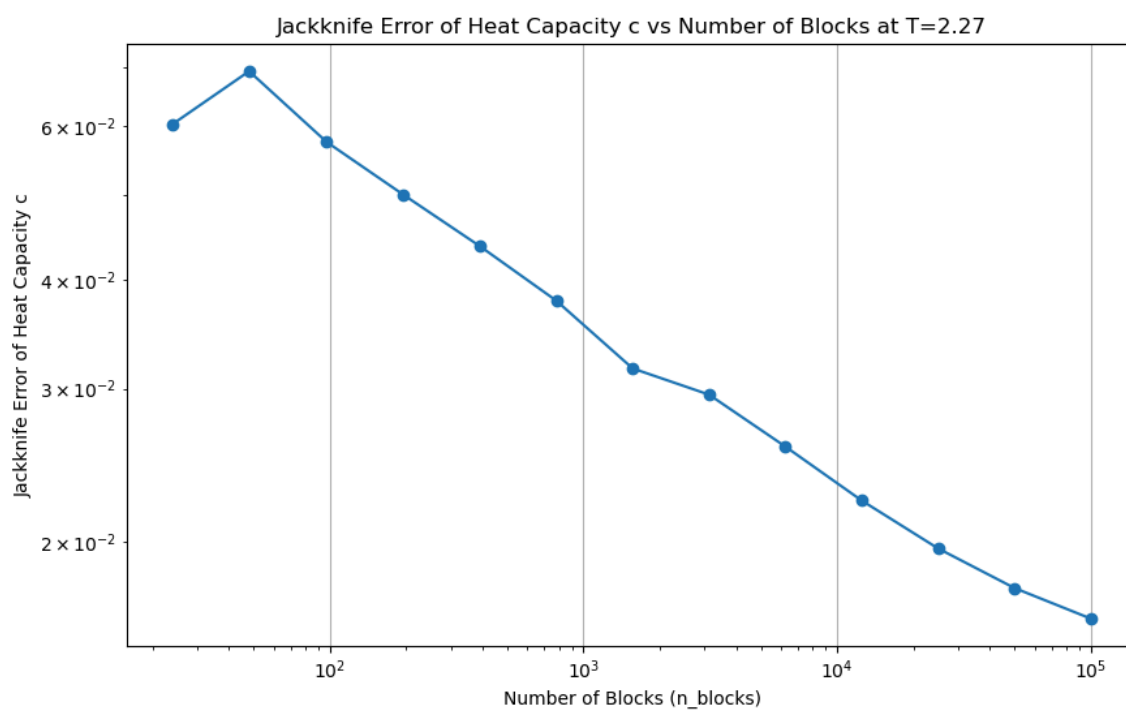
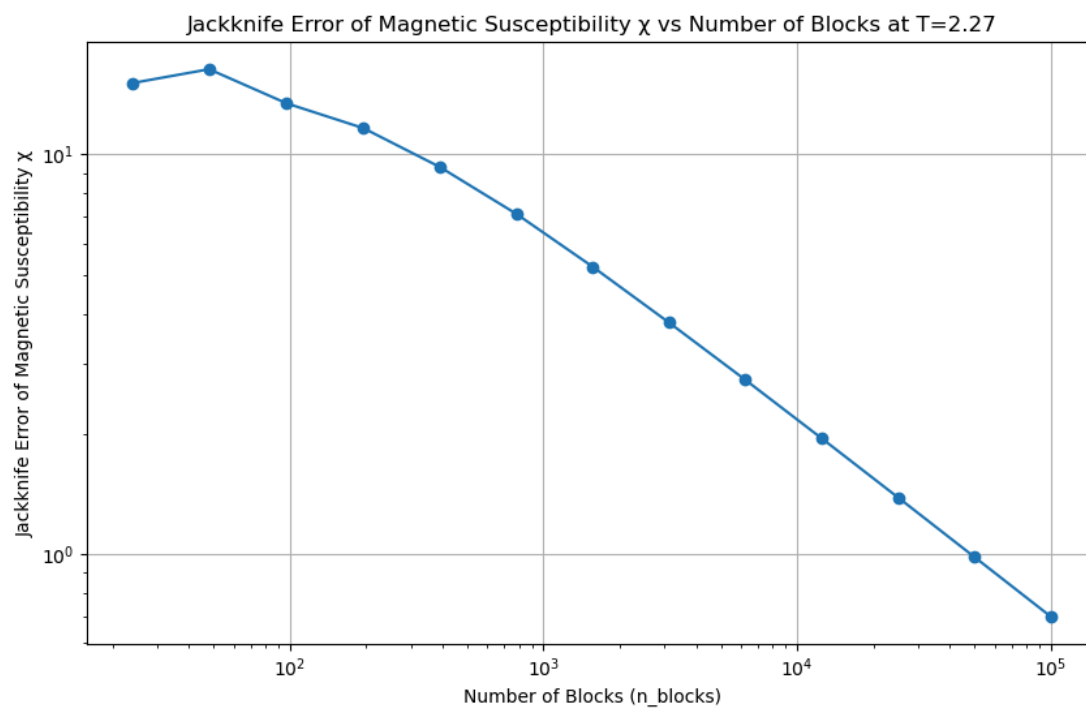
Interpretation

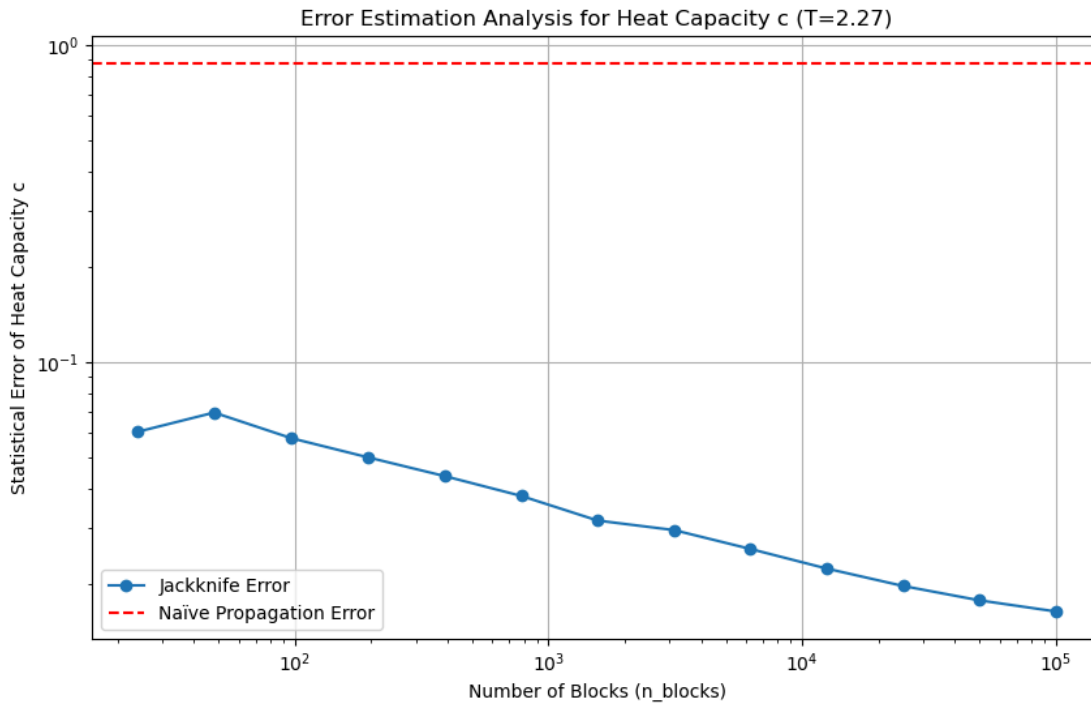
- At the low temperature ($T=2.0$), the error grows up to bin size 11 and plateaus, both for energy & magnetization. This state shows the lowest energy and highest magnetization.
- At the critical temperature ($T=2.27$), the convergence is prolonged. Mean absolute magnetization is around 0.5 as phase transitions can occur at this temperature.
- At the high temperature ($T=2.6$), the error rate is more constant throughout bin size, indicating fast decorrelation. The magnetization is low (averages towards zero) in this disordered state due to being above the critical temperature.

Note: The problem said to do a function of T , but this would have resulted in 16 plots (one for each bin size) in order to also represent the error at each bin. As a result, 3 graphs, one for each bin size, were created.

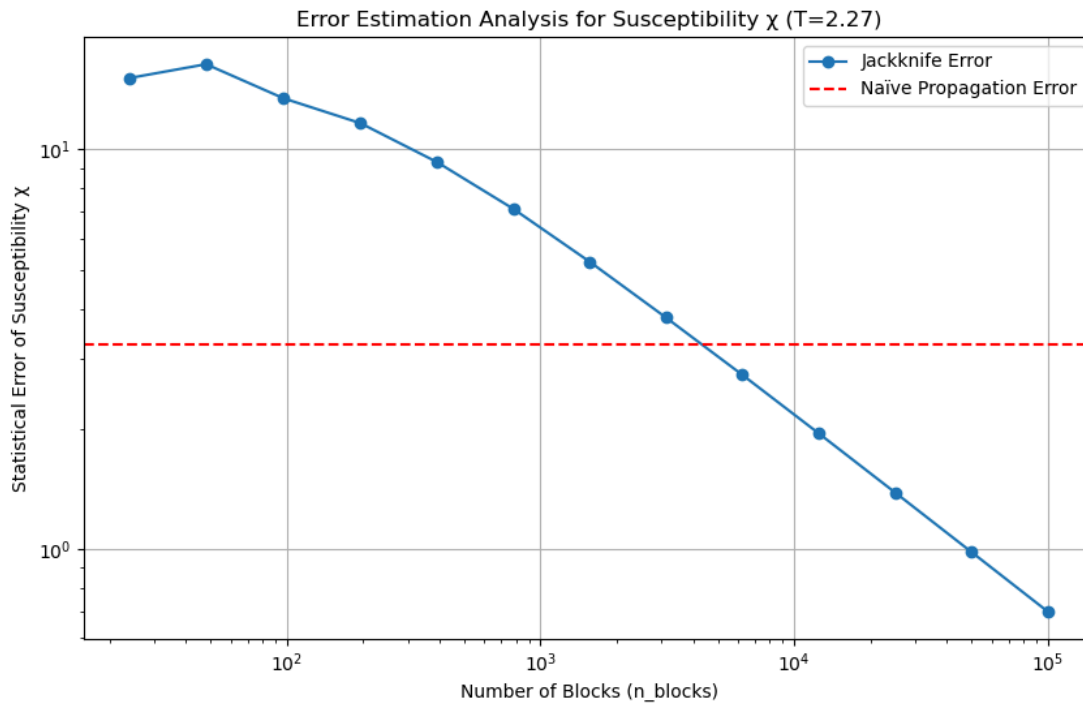


Optional task 1 & 2 - Jackknife Statistics for magnetic susceptibility, heat capacity per spin and Binder parameter





```
raw
Jackknife Plateau Error (approx): 0.060341 (using fewest blocks/largest block size)
Naïve Propagation Error: 0.878231
Ratio (Naïve / Jackknife): 14.55
```



```
raw
Jackknife Plateau Error (approx): 15.097334 (using fewest blocks/largest block size)
Naïve Propagation Error: 3.245069
Ratio (Naïve / Jackknife): 0.21
```

Heat Capacity

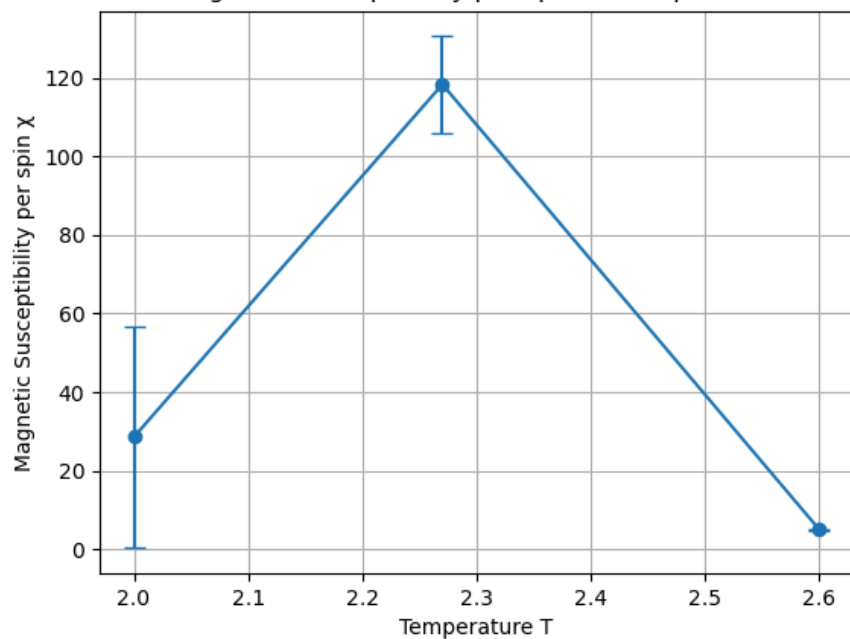
The Naïve error estimate is nearly 15 times larger than the Jackknife error for the largest bin size and over 50 times larger for the smallest bins. We get these much smaller Jackknife errors as a result of including the correlation between $\langle E^2 \rangle$ and $\langle E \rangle^2$. The fluctuations cancel out resulting in much smaller (true) value. In the Naïve Propagation we assume that $\langle E^2 \rangle$ and $\langle E \rangle$ fluctuate independently so we add their variances.

Susceptibility

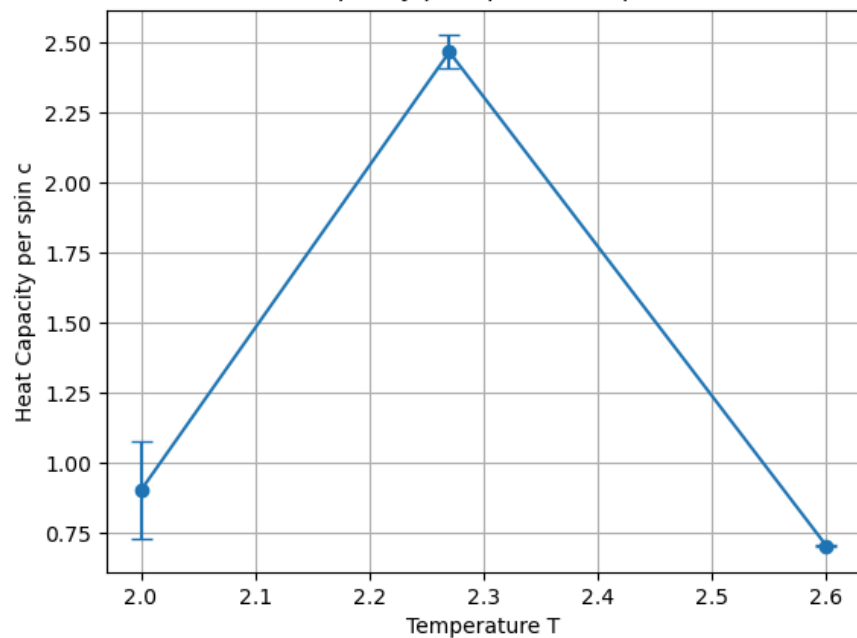
Here the Naïve error estimate is in the middle of the Jackknife errors range. It can be around 5 times smaller (for large bins) or 5 times larger (for small bins). It overestimates the error by ignoring the noise cancellation (covariance) between $\langle M^2 \rangle$ and $\langle |M| \rangle^2$, but simultaneously underestimates the error by ignoring the massive reduction in effective sample size caused by critical slowing down.

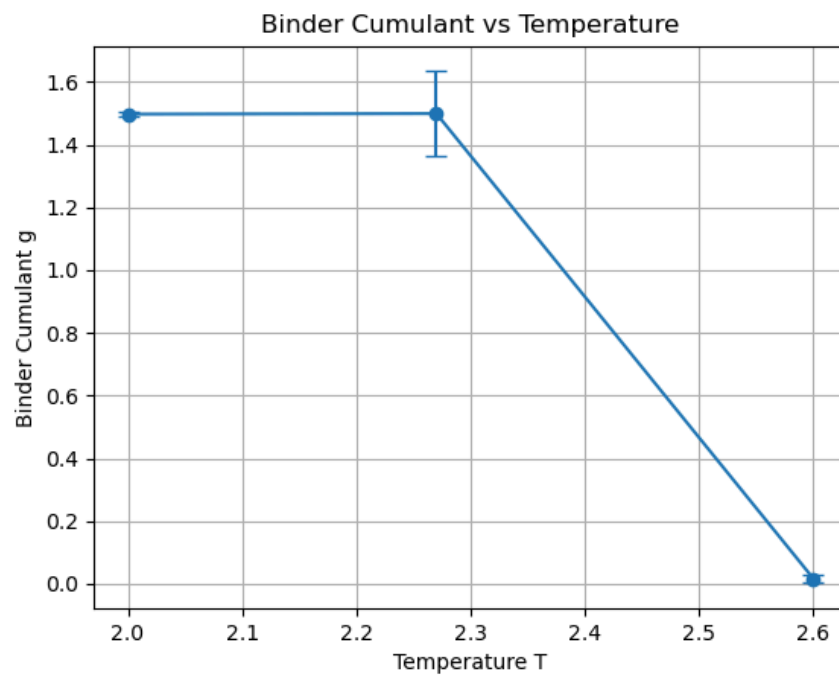
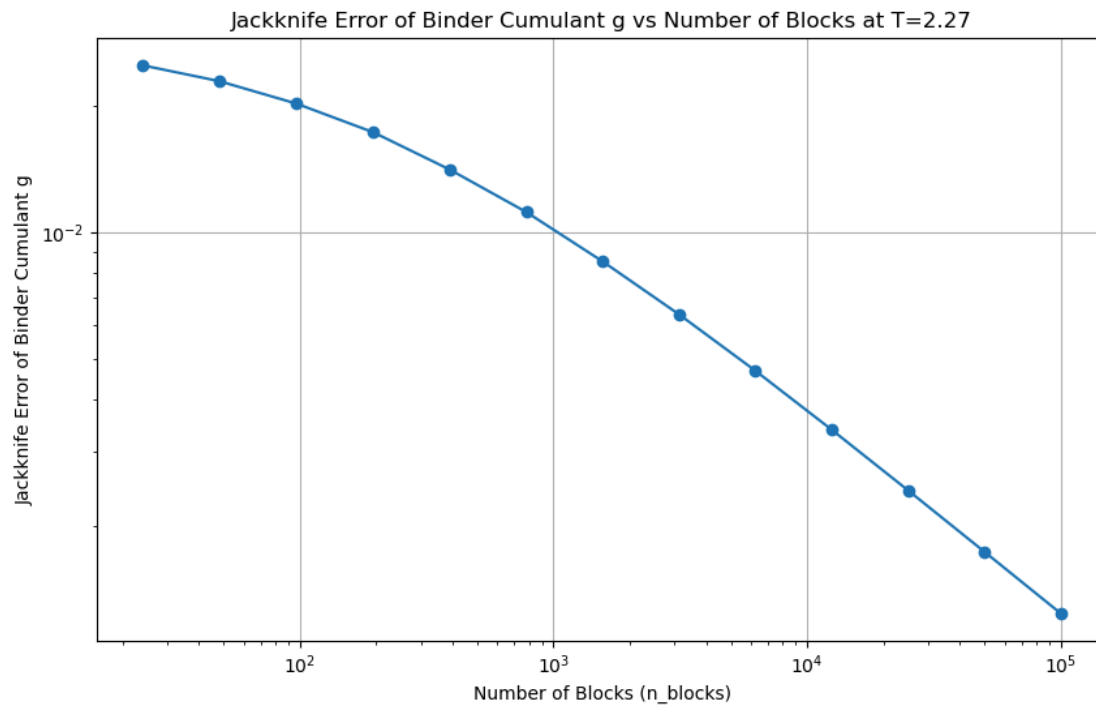
For small block sizes, the covariance effect dominates (Jackknife < Naïve). For large block sizes (the correct physical limit), the critical slowing down dominates, driving the true uncertainty well above the Naïve prediction (Jackknife > Naïve).

Magnetic Susceptibility per spin vs Temperature



Heat Capacity per spin c vs Temperature





How AI was used

1. After coming up with the neighbor calculations & metropolis step functions, AI (Gemini & Copilot) assisted with the numba just-in-time compiled functions so that the python code would be fast enough to run the Monte Carlo simulation, achieving $\sim 3.6e7$ spin flips per second.
2. The random number generator was proposed by AI and then tested for efficiency. Options were reviewed within Numpy documentation here: [<https://numpy.org/doc/stable/reference/random/generated/numpy.random.choice.html>]